

SQL PROJECT DOCUMENTATION

Project Title

AMAZON ANALYSIS DATABASE

Submitted By

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1. Introduction

The **Amazon Analysis Database** project is developed using SQL to store, manage, and analyze data related to an online shopping platform. The database contains information about customers, products, suppliers, orders, order details, and product reviews. The project demonstrates the use of SQL commands for database creation, data manipulation, data retrieval, constraints, joins, aggregate functions, grouping, sorting, and ranking.

2. Objectives

- Create an Amazon-style database.
 - Store customer, product, supplier, order, and review information.
 - Perform CRUD operations using SQL.
 - Generate business reports using SQL queries.
 - Analyze customer purchases and product performance.
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3. Database Used

Database Name: foodorders

4. Tables Used

- Customers
- Products
- Orders
- Order_Details
- Suppliers
- Reviews

5. SQL Operations Performed

Database Creation

```
CREATE DATABASE foodorders;
```

Display Records

```
SELECT * FROM foododer.customers;  
SELECT * FROM foododer.products;  
SELECT * FROM foododer.orders;  
SELECT * FROM foododer.order_details;  
SELECT * FROM foododer.reviews;  
SELECT * FROM foododer.suppliers;
```

Filtering Data

Retrieve customers from Texas.

```
SELECT *  
FROM foododer.customers  
WHERE State='Texas';
```

Retrieve all products under the Fruits category.

```
SELECT *  
FROM foododer.products  
WHERE Category='Fruits';
```

Applying Constraints

```
ALTER TABLE foododer.customers  
ADD CONSTRAINT unique_name  
UNIQUE(Name);
```

This ensures that duplicate customer names are not allowed.

Inserting Records

New product records were inserted using the INSERT statement.

```
INSERT INTO foododer.products(...)
VALUES (...);
```

Aggregate Functions

Average rating for each product.

```
SELECT ProductID,
AVG(Rating) AS AverageRating
FROM Reviews
GROUP BY ProductID;
```

GROUP BY

Total quantity ordered for each product category.

```
SELECT p.Category,
SUM(o.Quantity) AS Orders_Placed
FROM Products p
JOIN Order_Details o
ON p.ProductID=o.ProductID
GROUP BY p.Category;
```

ORDER BY

Display categories based on highest number of orders.

```
ORDER BY Orders_Placed DESC;
```

LIMIT

Display the Top 3 ordered categories.

```
LIMIT 3;
```

JOIN Operations

Joining Products and Order_Details tables to generate sales reports.

```
SELECT p.Category,  
SUM(o.Quantity)  
FROM Products p  
JOIN Order_Details o  
ON p.ProductID=o.ProductID  
GROUP BY p.Category;
```

6. Business Analysis Performed

- Customer filtering by location.
 - Product category analysis.
 - Average product rating calculation.
 - Most ordered product categories.
 - Product insertion.
 - Data validation using constraints.
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7. Results

The SQL queries successfully retrieved meaningful information from the database. Aggregate functions and JOIN operations provided insights into customer purchasing behavior, product popularity, and product ratings.

8. Conclusion

The **Amazon Analysis Database** project demonstrates the practical application of SQL for database management and business analysis. The project covers database creation, data retrieval, insertion, constraints, joins, grouping, sorting, and reporting techniques. These SQL operations are essential for managing real-world e-commerce databases efficiently.